Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Kinematics 4: Calculations — ANSWERS

You are not being asked to work anything out for yourself on the blue panels.

Every formula is written out. Every letter is labelled. Every rearrangement is already done for you — you never have to move a letter across an equals sign on this sheet. Read the blue panel, read the worked example beside it, then copy that shape into the questions. When you get stuck, look *up the page*, not inside your head.

The answer key is on the class page. Do the question first, then check it. Checking an answer you have not attempted teaches you nothing.

The four-step method

use this on every calculation

Every calculation question on this sheet — and in the exam — is done the same four ways. Write all four steps down every time, even when you can see the answer.

Step 1	**WRITE**	what the question gives you, with units.
Step 2	**CHOOSE**	the formula, in the form that already has the letter you want on its own.
Step 3	**PUT IN**	the numbers, in place of the letters.
Step 4	**ANSWER**	the number *and its unit*. A number with no unit scores zero.

The units you need on this sheet. Nothing else appears.

Quantity	Symbol	Unit	Short
distance, displacement	s	metres	m
speed, velocity	v	metres per second	m/s
time	t	seconds	s
acceleration	a	metres per second squared	m/s^2

Section 1 — Distance and displacement

no formula needed

Distance is how far you actually travelled. Add up every bit of the journey. It has **no direction**. It is a **scalar**.

Displacement is the straight line from where you *started* to where you *finished*, plus the direction. It is a **vector**.

The short version: distance is the road you drove. Displacement is the arrow from your house to the shop.

If you finish where you started, your displacement is zero — no matter how far you walked.

Some journeys on this sheet turn a corner — and you still will not need any trigonometry. Every question is built so that the legs pointing opposite ways cancel each other out exactly. You will never have to draw a sloping line, and you will never need Pythagoras, sine or cosine. Adding and subtracting is the whole of it.

Worked example — copy this shape

A student walks **6 m east**, then turns round and walks **2 m west**.

Distance $= 6 + 2 =$ **8 m** (add up everything you walked)
Displacement $= 6 - 2 =$ **4 m east** (where you ended up, from the start, *and* the direction)

1. A student walks **10 m east**, then **4 m west**.

Distance **14 m** Displacement **6 m east**

2. A student walks **3 m east**, then **3 m west**.

Distance **6 m** Displacement **0 m**

3. A runner runs **100 m** straight down a track and stops.

Distance **100 m** Displacement **100 m, down the track**

4. A cyclist rides **12 m east**, then a further **5 m east**.

Distance **17 m** Displacement **17 m east**

5. A runner completes one full lap of a **400 m** track, finishing where they started.

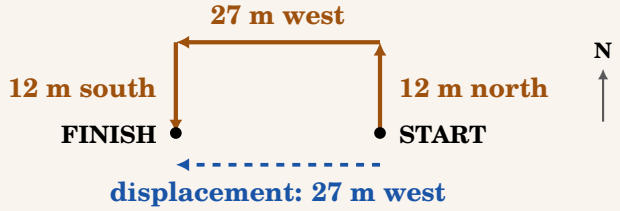
Distance **400 m** Displacement **0 m**

Section 1, continued — a journey that turns a corner

Same two questions, same two rules. Draw the journey first — every time.

Worked example — copy this shape

A student walks **12 m north**, then **27 m west**, then **12 m south**.



Distance $= 12 + 27 + 12 =$ **51 m** (add up every leg you walked)

Displacement: the 12 m north and the 12 m south *cancel* — you went up and came straight back down again, so you end up level with where you started. All that is left is the 27 m west.

Displacement = 27 m west

Notice how far apart the two answers are: 51 m walked, but only 27 m from start to finish. That gap is the whole reason physics needs two different words.

6. A student walks **8 m north**, then **15 m east**, then **8 m south**.

Distance **31 m** Displacement **15 m east**

The 8 north and the 8 south cancel. $8 + 15 + 8 = 31$ *m walked, but only 15 m from start to finish.*

sketch your journey here, and label each leg — it does not need to be to scale

7. A student walks **5 m west**, then **20 m north**, then **5 m east**.

Distance **30 m** Displacement **20 m north**

This time it is the 5 west and the 5 east that cancel, so the leftover leg is the 20 m north. The cancelling pair is not always the first and last numbers — it is always the two that point opposite ways.

sketch your journey here, and label each leg — it does not need to be to scale

distance travelled $=$ $\boxed{s = v \cdot t}$ $=$ speed $\times$ time

That box is equation 4 on the AQA formula sheet. It is worth knowing what the real thing looks like, because it is what you will be handed in the exam.

One difference, and it is worth a look. AQA prints it as $s = v\,t$ — **no symbol at all between the v and the t.** Two letters written side by side means **multiply**. Every equation on their sheet is written that way ($W = m\,g$, $F = m\,a$). **This sheet puts the dot in** so there is no doubt what you are being asked to do.

And the units come straight out of it. Metres divided by seconds *is* metres per second, **m/s**.

Speed and velocity — the same distinction you have already met.

	no direction (scalar)	**has a direction (vector)**
how far	distance	displacement
how fast	**speed**	**velocity**

Speed is to velocity exactly what distance is to displacement. The number is worked out the same way both times — the only difference is whether you say which way at the end. So there is one formula here, not two.

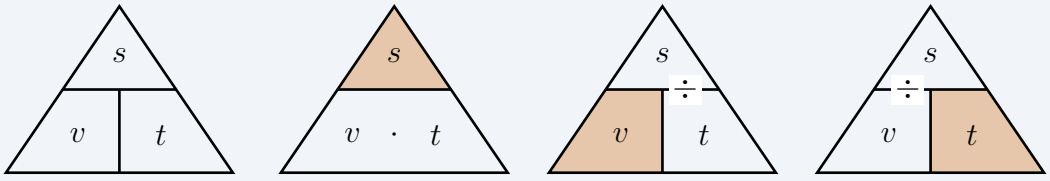
This is why the formula sheet only gives you one. AQA's equation 4 says *distance* and *speed*. Put a direction on each and the very same equation gives you *displacement* and *velocity*. Same letters, same arithmetic — you just say which way at the end.

All three forms. You never have to rearrange anything — pick the line with the letter you want on the left.

$v = s \div t$	looking for **velocity**? use this one
$s = v \cdot t$	looking for **displacement**? use this one
$t = s \div v$	looking for **time**? use this one

And when you do see a fraction, the line just means divide. $\frac{s}{t}$ and $s \div t$ are the same instruction written two ways. Equation 4 is written as a multiplication, but equation 5 in the next section is written as a fraction — so you will meet both.

Triangle math — if you would rather not read the three lines above. Put the letters in the triangle. **Shade the one you are looking for**, and what is left in the other two boxes is what you do.



the triangle **shade s:** $s = v \cdot t$ **shade v:** $v = s \div t$ **shade t:** $t = s \div v$

The top of the triangle is always the one on its own in the formula — here that is s. The two at the bottom are multiplied together.

The minus sign is a direction, not a mistake. If we agree that east is positive, then a velocity of -3 m/s simply means 3 m/s *west*. Nothing is going backwards in time and nothing is broken.

Worked example — copy this shape

A cart moves **12 m west** in **4 s**. What is its velocity?

WRITE $s = 12$ m west $t = 4$ s $v = ?$
CHOOSE $v = s \div t$
PUT IN $v = 12 \div 4$
ANSWER $v =$ **3 m/s west** (or $-$**3** m/s, if east is positive)

8. A car travels at **15 m/s east** for **4 s**. What is its displacement? **60 m east**

9. A runner runs **60 m east** in **40 s**. What is their velocity? **1.5 m/s east**

10. A cart moves **9 m west** in **3 s**. Taking east as positive, what is its velocity? -3 **m/s**

11. A trolley has a displacement of -10 **m** in **2 s**. What is its velocity? -5 **m/s**

12. A car travels **300 m east** at a velocity of **20 m/s**. How long does it take? **15 s**

13. One student walks at **2 m/s east**. Another walks at **2 m/s west**.

Do they have the same **speed**? **Yes** Do they have the same **velocity**? **No**

Same number, opposite directions. Speed ignores direction; velocity does not. This is the whole scalar / vector idea in one question.

Working space — show the four steps for at least two of the questions above.

$$\text{acceleration} = \boxed{a = \frac{\Delta v}{t}} = \frac{\text{change in velocity}}{\text{time taken}}$$

That box is equation 5 on the AQA formula sheet, copied exactly — words and all.

Only the top of the fraction has a Δ. The triangle means **change in**. The bottom is just t, the **time taken** — no triangle.

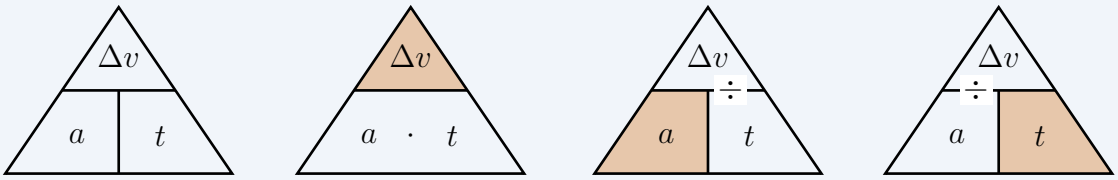
You work the top out like this:

$$\Delta v = \textbf{final velocity} - \textbf{initial velocity} \qquad \text{or} \qquad \Delta v = v - u$$

Quantity	Symbol	Meaning	Unit
initial velocity	u	the one it **started** at	m/s
final velocity	v	the one it **ended** at	m/s

Final minus initial, in that order. The other way round and every slowing-down answer comes out with the wrong sign.

Triangle math again. Δv goes on top, *not* a. Shade what you want.



the triangle **shade Δv:** $\Delta v = a \cdot t$ **shade a:** $a = \Delta v \div t$ **shade t:** $t = \Delta v \div a$

Watch this one. It is tempting to put a on top, because acceleration is what the formula is called. **Whatever is on its own when there is no dividing line goes on top** — and that is Δv.

A negative answer means it is slowing down. That is called **deceleration**. It is not an error — do not chop the minus sign off.

Worked example — copy this shape

A car speeds up from **rest** to **12 m/s** in **4 s**. What is its acceleration?

WRITE $u = 0$ m/s ("from rest" means the initial velocity is zero) $v = 12$ m/s $t = 4$ s
CHOOSE $a = \Delta v \div t$, where $\Delta v = v - u$
PUT IN $\Delta v = 12 - 0 = 12$ m/s, so $a = 12 \div 4$
ANSWER $a = \mathbf{3\ m/s^2}$

14. A car speeds up from **rest** to **20 m/s** in **5 s**. What is its acceleration? **4 m/s^2**

15. A cyclist speeds up from **4 m/s** to **16 m/s** in **6 s**. What is their acceleration? **2 m/s^2**

16. A car slows from **30 m/s** to **rest** in **10 s**. What is its acceleration? **-3 m/s^2**

$\Delta v = 0 - 30 = -30\ m/s$, so $a = -30 \div 10 = -3\ m/s^2$. Finish minus start, in that order. Writing $30 - 0$ gives $+3$ and loses the mark.

17. A trolley slows from **12 m/s** to **4 m/s** in **4 s**. What is its acceleration? **-2 m/s^2**

$\Delta v = 4 - 12 = -8\ m/s$, so $a = -8 \div 4 = -2\ m/s^2$. It is still moving forwards — the minus says it is slowing, not reversing.

18. A skateboarder starts from **rest** and accelerates at **2 m/s^2** for **5 s**.

What velocity do they reach? **10 m/s**

Working space — show the four steps for at least two of the questions above.

Before you hand this in, check three things.

1. Has every answer got a **unit** after it? **2.** Have you written all **four steps**, not just the answer?
3. If an answer came out **negative**, did you keep the minus sign?